Cross-lingual intervention for low-resource Machine Translation

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Bridging the Language Gaps in Large Language Models with Inference-Time Cross-Lingual Intervention

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Motivation

- Large language models achieve strong overall performance
- However, they remain **heavily English-centric**

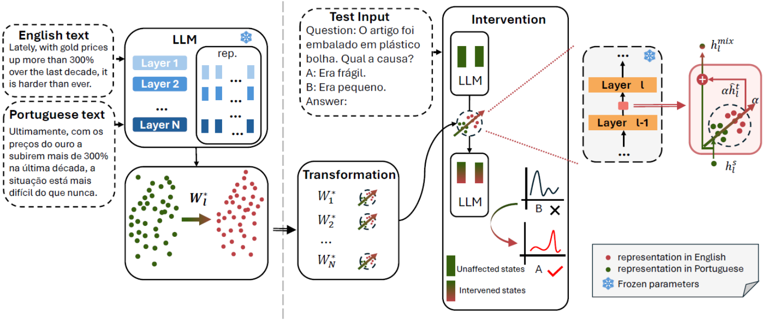
- Significant performance gaps for **low-resource languages**
- Fine-tuning or retraining multilingual models is **computationally expensive**

Research Question

Can we reduce cross-lingual performance gaps

without retraining the model ?

Method Overview



(a) Learning the Cross-Lingual Alignment
(b) Inference-Time Transformation

We propose **INCLINE** : Inference-Time Cross-Lingual Intervention

- Aligns internal representations across languages
- Transfers knowledge from high- to low-performing languages
- Applied only at inference time with no parameter updates

Stage 1 : Learning Cross-Lingual Alignment

Goal : Map source-language representations to target-language space.

For each layer l, we learn a linear mapping W_l :

$$W_l^* = \arg\min_{W_l} \sum_{i=1}^{N} \left\| W_l h_{i,l}^s - h_{i,l}^t \right\|^2$$

- $h_{i,l}^s$: hidden state (source language)
- $h_{i,l}^t$: hidden state (target language)
- Alignment learned offline using parallel data

Stage 2 : Inference-Time Intervention

Given a test input in the source language :

Projection

$$\hat{h}^{t}_{q,l} = W^{*}_{l} h^{s}_{q,l}$$

Intervention

$$h^{\mathrm{mix}}_{q,l} = h^{s}_{q,l} + \alpha \hat{h}^{t}_{q,l}$$

Where :

- l : layer index
- h^{s}/h^{t} : hidden states of source / target language
- α : intervention strength

Limitations

- **Over-reliance on XStoryCloze :** Most experiments focus on XStoryCloze (a discriminative task), while limited discussion is provided on generative tasks like Translation.
- **Focus on High-Resource (X) to English :** While the method has potential for low-resource languages, only high-resource (X) to (English) translation was considered.
- **Reliance on BLEU :** Evaluation is limited to BLEU scores even though LLMs tend to paraphrase significantly.

Improved Methodology

Proposed Improvement : Consistency
INCLINE trains W for direct projection ($Wh^s \approx h^t$) but uses it for mixing ($h^s + \alpha Wh^s$) during inference, creating an objective mismatch.

We modify the training objective to directly optimize the post-intervention vector, ensuring W learns the specific shift needed given the mixing factor α.

New Training Objective :

$$W_l^* = \arg\min_{W_l} \sum_{i=1}^{N} \left\| \left(h_{i,l}^s + \alpha W_l h_{i,l}^s \right) - h_{i,l}^t \right\|^2$$

Evaluated Approaches

- **No intervention** uses the quantized base model for translation, with no alignment or model modification.
- **Intervention** applies the learned alignment matrices at inference time, modifying hidden states using a fixed intervention strength (referred to as INCLINE).
- **Consistent intervention** applies the proposed improvement in training the alignment matrices and applies these alignment matrices in the same way during inference (referred to as Improved INCLINE).
- **LoRA** trains low-rank adapters on the same FLORES translation data. Since both LoRA fine-tuning and INCLINE adds additional intermediate weights, by comparing both, we gauge into the effectiveness of the indirect INCLINE objective against the more apparent token prediction loss.

Implementation details

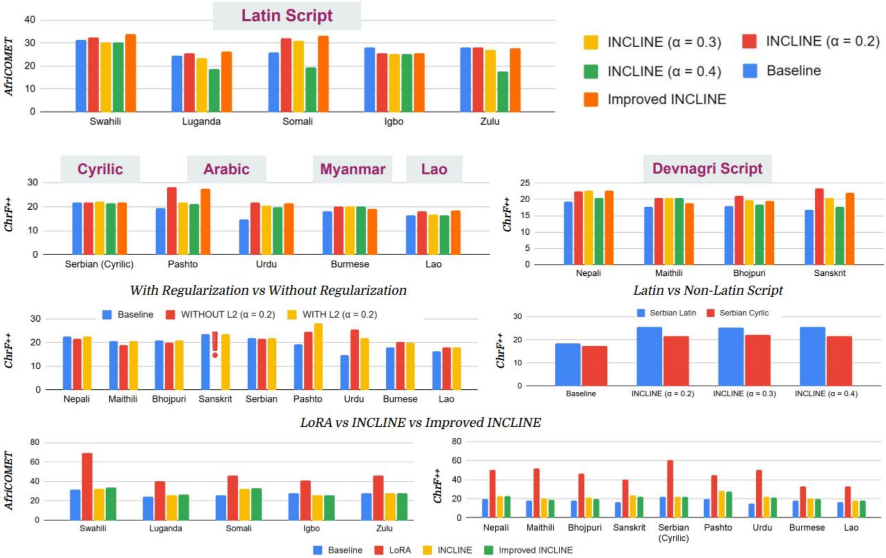
- We start with Llama-3-8B-Instruct as our base LLM.
- We focus exclusively on the **low-resource translation** task.
- All experiments use quantized model for viable compute.
- Sentence representations are extracted from the **last token** of both source and target sentences.
- We use **L2 regularization** since the system is highly overparameterized.
- Translation quality is evaluated using **BLEU** and **ChrF++**. For African languages, we additionally report **AfriCOMET**.

Our experiments

Questions we seek to answer :

- Whether INCLINE works well for low resource languages
- Whether performance of INCLINE is dependent on the script of the high resource language
- Whether tokenization density affects the performance.

Our experiments



Takeaway

- **INCLINE** performs well even in severe **low-resource settings**.
- **LoRA** significantly outperforms **INCLINE**, but **INCLINE** offers a much quicker and more efficient way to improve **base model**.
- Improvement is consistent across **scripts**, though better for **Latin scripts** (e.g., Serbian Latin vs. Cyrillic).
- **INCLINE** shows improvement even for languages with very **sparse tokenization** (Lao & Burmese).
- Our proposed **INCLINE improvement** achieves better performance for **African Languages** while matching performance for others.
- The **regularized objective** is stable, unlike the **non-regularized approach** which failed on Sanskrit.

References

- Wang, Jiayi et al. "AfriMTE and AfriCOMET : Enhancing COMET to Embrace Under-resourced African Languages." North American Chapter of the Association for Computational Linguistics (2023).
- Popović, Maja. "chrF++ : words helping character n-grams." Proceedings of the second conference on machine translation. 2017.
- AI4Bharat et al. "IndicTrans2 : Towards High-Quality and Accessible Machine Translation Models for all 22 Scheduled Indian Languages." ArXiv abs/2305.16307 (2023).
- Wang, Weixuan, et al. "Bridging the language gaps in large language models with inference-time cross-lingual intervention." Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1 : Long Papers). 2025.
- team, Nllb et al. "No Language Left Behind : Scaling Human-Centered Machine Translation." ArXiv abs/2207.04672 (2022).
- Hu, J. Edward et al. "LoRA : Low-Rank Adaptation of Large Language Models." ArXiv abs/2106.09685 (2021).